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Description

Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

My program is the airplane game that a player controls airplane which must escape from enemies' attacks. The only way to move the airplane is typing some keys: w, s, a, d.
The airplane can move in both horizontal and vertical way, while the attacks will move in vertical way.
The ultimate goal in this game is not avoiding all the attacks but collecting scores which will be the key to win the game. Player should score at least 100 to win. There will be three levels, and each level will represent the difficulty of game. After finishing each game through score 100, player will reveice a winner label that represents your winning. If you die during the game, you must restart the game from zero score.
Additionally, there will be no limits with moving airplane outside of window unless hit by attacks.

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?

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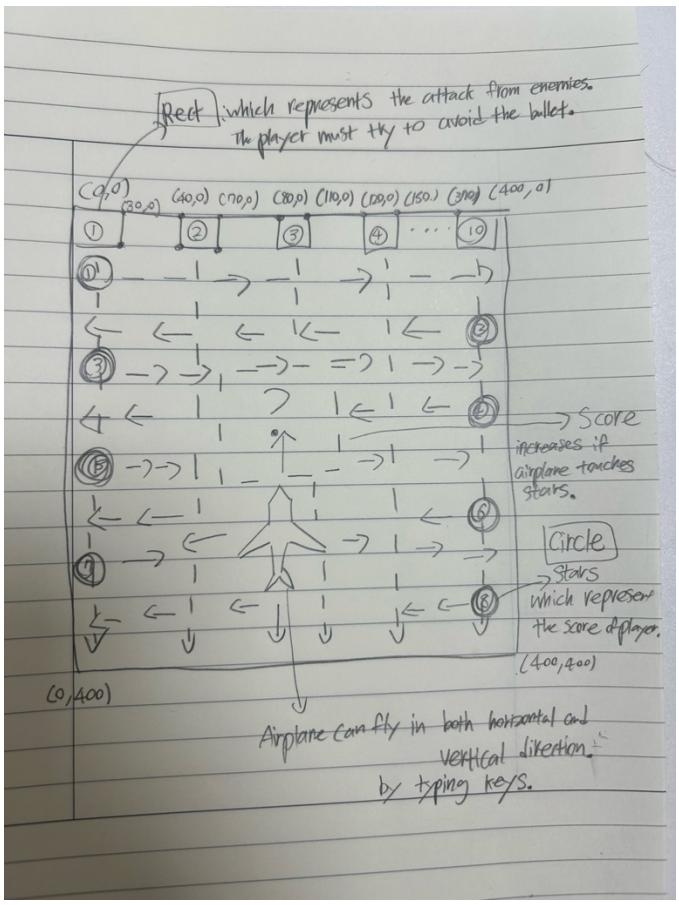

Unit	Introduced Concept
1	Creating drawings, shapes, colors, opacity
2	Functions, onMousePress and onMouseRelease, and shape properties
3	onMouseMove and onMouseDrag, conditionals, helper functions
4	elif statements, onKeyPress and onKeyRelease, shape methods

Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

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Concept	Purpose
Elif statements	I plan on using Elif statements to create different levels of game, different movements of airplane, and overall various scenarios at the same time. Even though the result of hitting by attacks will be always same, which is death in game, I plan to count the attacks one by one because each has slightly different speed moving.
onKeyPress	I plan on using onKeyPress to create the movements of airplane. By using onKeyPress, a player can move the airplane in order to avoid the attacks and to score.
Shape methods (hitsShape, toFront, toBack, energy)	I first plan on using toFront and toBack to fix some wrong orders of code, such as cursor behind the background and new background covered by old background. I second plan on using hitsShape to make the systems in game. For example, if airplane hit by attacks, game is over. On the other hand, if airplane touches the white circles, which represent the actual score of game, the current score in that level of game will increase based on the number of interacting with the circles. Lastly, I plan on using energy to build the system of win, which is scoring enough that brings win.
onMouseMove	I plan on using onMouseMove to create a cursor for player so he can choose the level of game himself.
Conditionals if	I first plan on using conditionals to create a pattern of attacks that continues repeatedly until a player is died and a pattern of stars in movement. I secondly plan on using conditionals to create the system that calculates how many points the player does have currently as he gets point.
Shapes(label)	I plan on using shapes to create two types labels: one for re-try and another one for winning label.
onMousePress	I plan on using onMousePress to determine the levels of game. If a player clicks level 1 shape, it will allow him to enter level 1 game.
Shape properities(+=)	I plan on using shapes properties to make attacks, stars and airplane move at the same time. By using shape properties, I can create the environment that all different shapes run together.

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

My problem during the coding was that I tried to create the game by using only one if function. Therefore, I could not put each specific detail and code for different scenarios and confused where to fix and how to restart building a game. Everything did not work at the same time. Therefore, I brought the if-elif statements in order to divide each important part and put details differently depending on the type of scenarios. At the end of coding, I realized that my program actually needed a lot of coding details much more than I expected and if-elif statements were extremely helpful for building my program.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
Elif statement	64-84 92-810 (broadly) 93-119 162-190 239-267 272-298 341-369 417-445 452-478 538-566 598-626 632-658 720-748 780-808	Elif statement is a type of conditionals in python used for calling different scenarios by shaping one group	Elif statement is used to call different situations at the same time. Firstly, it is used to divide the level of game into three. Secondly, it is used to create various horizontal and vertical movements of airplane and the velocities of attacks based on the keys pressed. Therefore, I needed elif statement to make different responses for each key pressed. Thirdly, it is used to create the system of score that counts as the airplane touches with stars. Since stars count one by one, I needed to accumulate individualy through elif statement.
onKeyPress	90-810(broadly)	onKeyPress is a type of key events used for calling when key is pressed in	I used onKeyPress to make the airplane and other things such as attacks and stars move. When key “w” is pressed, the

		python.	airplane moves up in vertical way, and the attacks move down, while stars move in horizontal way.
Shape methods (hitsShape, toFront, toBack, energy,app.stop())	64-84 162-187 194-206 239-264 341-383 417-442 522-563 598-623 704-745 780-805	1. hitsShape is a type of shape methods used to run if one shape hits by other shape 2. toFront is a type of shape methods used to bring a shape in front of other shapes 3. toBack is a type of shape methods used to push a shape to the back of other shapes 4. energy is a type of shape methods referring to the repeated number of particalr act in python 5. app.stop is a type of shape methods used to call when one wants to stop running code immediately	I used hitsShape to determine whether the airplane was hit by attacks or stars. If there was an interaction between airplane and attacks, then the program will stop immediately through the code 'app.stop()', which refers to the lose in that game. If airplane touches stars, then there will be increase in values of score, which refers to the only key to win the game. I used toFront and toBack to correct some wrong orders of code, such as new background covered by old background. Energy was used to calculate the score of player in the game. If a player reaches 100 score, then the game will label the winning.
onMouseMove	58-60	onMouseMove is called every time the mouse is moved and not pressed	I used onMouseMove to move cursor. In the game, mouse is used to choose the level of game.
Conditonals	130-162 194-239 308-337 371-414 488-534 568-596 668-716 750-778	Conditionals, especially if statements, are a type of codes that run only if some test is true.	I used conditonals to make a repeating pattern of attacks and stars in the game. Attacks and stars move normally, but when they ran out of window, they re-positioned in an initial positon through conditonals. Addtionally, by using condtionals, I built particular responses, such as app.stop if the airplane was hit by attacks and label that shows winning at the end of game if the player reached over 100 score.

Shapes(label)	6,7 164-189 196-208 371-385 447 524-536 628 706-718 810	Label is a type of shapes that can be shown in terms of letter in window (400x400).	I used shapes (label) to sign if airplane was hit by attacks or player won the game. Additionally, I used label to show the calculated values of score in the game. If the value of label reaches out 100, then it means the player won the game. If the airplane interacts with stars, the value of score increase one by one.
onMousePress	63-84	onMousePress is a type of code that rusn if mouse is pressed.	I used onMousePress to determine which level of game the player have chosen to enter.
Shape properties	94-128 210-223 272-306 452-486 632-666	Changing properties is used to add or substract some values in python.	I used shape properties to move the airplane in both horizontal and vertical ways, to move the stars in horizontal, and to move the attacks in vertical way.

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

For code changes aspect, I will try to abridge the full length of my codes because I could not find which line went to wrong and was confushed where to fix. To shorten the length of code next time, I think I have to learn new codes that can duplicate original codes even in different if situations.
For process adjustments, I think there were a lot of deviations during even building a game. Even though these thoughts delayed to complete the program, I think these deviations rather led me to create a fancy game now. But, in the future, I will narrow down the direction of coding and build funnier game next time. For resource utilization, I tried to utilize the exampler of creative task given by my teacher and review all the contexts we have covered in order to know the exact purpose and usage of codes. Through

the inforamation, I could find out what is the difference between my program and others. For next time, I will try to look up some other online resources additionally in order to expand my work in positive way. For time management aspect, I think I did not handle very effectively since I was struggled to decide what game I should make. Therefore, I will try to work on time management as well in order to maximaize my coding work and make a funny game next time by creating a schedule of building program before the due day of creative task.

Scroll down for submission checklist.
Pre-Submission Checklist for Creative Tasks

Program Design

- [ ] Described what your program will do clearly.
- [ ] Incorporated at least two concepts from the current unit.
- [ ] Added annotations to explain colors, movements, and user interactions.

Canvas Design

- [ ] Sketched your canvas layout.
- [ ] Labeled any animation or user interaction on the canvas sketch.
- [ ] Detailed dominant colors and their importance.
- [ ] Described any animations or movements.

Code Design

- [ ] Listed at least four to six coding concepts or properties you plan to use.
- [ ] Explained why each concept or property is essential for your program.

Program Development and Coding

- [ ] Finished coding the program.
- [ ] Debugged and tested the program.
- [ ] Checked if the program meets the design criteria.

Difficulties and Opportunities

- [ ] Described a difficulty faced or an opportunity realized in detail.
- [ ] Explained the steps taken to resolve the difficulty or seize the opportunity.

Finished Project Code Outline

- [ ] Listed only one instance of each important coding concept used.
- [ ] Provided the line numbers for each coding concept.
- [ ] Included a brief definition for each coding concept.
- [ ] Highlighted the aspect you are most proud of.

Future Goals

- [ ] Discussed what specific changes you'd make if given more time.
- [ ] Reflected on your approach and planning for the project.
- [ ] Considered resource utilization and time management.

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Final Check

- [ ] Revisited the rubric to make sure you meet all the criteria.
- [ ] Made sure your document is well-organized and free of grammatical errors.
- [ ] Ready to submit the project and this document.

By ticking off each item on this checklist, you ensure that you have met all the requirements and can confidently submit your work for evaluation.